

Understanding users' AI manipulation intention: An empirical investigation of the antecedents in the context of AI recommendation algorithms

Taeyoung Kim, Il Im^{*}

School of Business, Yonsei University, Seoul, South Korea

ARTICLE INFO

Keywords:

AI manipulation behavior
Interactivity
Persuasion knowledge model
Recommendation system
Strategic manipulation behavior

ABSTRACT

This study examines antecedents that drive platform users to manipulate artificial intelligence (AI) recommendation algorithms. Based on the persuasion knowledge model (PKM), survey data collected from YouTube and Instagram users reveal that AI manipulation intentions are positively affected by persuasion knowledge about AI and perceived interactivity. Perceived interactivity is associated with higher perceived benefits and lower perceived costs of AI manipulation, consequently affecting manipulation intentions. A multivariate analysis of variance shows variations in intentions to use different types of AI manipulation behaviors among users with varying levels of persuasion knowledge. The research contributes to the PKM and AI-human interaction literature.

1. Introduction

Due to advanced artificial intelligence (AI) such as recommendation algorithms and ChatGPT, interactions with AI are becoming more frequent and intensive. Unlike traditional systems, AI learns continuously through user interactions [1]. A recent survey showed that 27 % of Americans engage with AI several times daily, and 60 % of American adults can identify at least three of the six representative examples of AI [2].

AI recommendation algorithms are unique in that users can strategically manipulate AI to obtain desired outcomes [3,4]. Specifically, on platforms such as YouTube and Instagram, online users can intentionally click on “like” buttons, even when they lack particular interest in the content, so that AI algorithms will make recommendations that match their tastes in the future [5,6]. In this study, we define this phenomenon as AI manipulation behaviors where users intentionally perform signaling actions to directly and indirectly cause AI to progressively produce more desirable results. Signaling actions can include clicking on content or pressing “like” and “dislike” buttons. AI manipulation behavior diverges from ordinary AI usage behavior where users press “like” buttons because they genuinely like and prefer specific content, without intentions to influence recommendations. Instead, AI manipulation behavior occurs when users specifically intend to alter AI outputs to receive better suggestions and have an expectation that their behaviors will lead to more desirable future outcomes, such as more tailored

shopping advice from Amazon and more personalized viewing endorsements from Netflix [1].

Similar concepts to AI manipulation have been explored across various disciplines, including strategic classification, counterfactual explanation, adversarial attacks, algorithmic rhetoric, and steering behavior. However, some of these concepts have some differences or ambiguities with AI manipulation. Strategic classification refers to a phenomenon in which users strategically report manipulated features to deployed classification algorithms (classifiers) in order to be categorized into a desired class [4,7]. Strategic classification behaviors and AI manipulation behaviors are similar in that both are intended to influence algorithms, but they differ in their ultimate goals. In strategic classification, users manipulate algorithms to achieve immediate positive classification outcomes such as loan approvals [4,7,8]. In contrast, AI manipulation behaviors are interactions with AI intended to change longer-term results. On the other hand, counterfactual explanation, adversarial attacks, algorithmic rhetoric, and steering behavior all aim to influence long-term outcomes similarly to AI manipulation, but each focuses on distinct areas: interpretable AI, cyberattacks, means of persuasion to influence public opinion and algorithm optimization. Specifically, counterfactual explanation is a technique for enhancing AI transparency by explaining how to change the outcome of an AI to users [9]. Adversarial attacks mean cyberattacks where manipulated data is fed to AI systems to cause malfunction [10]. Algorithmic rhetoric refers to persuading others by strategically manipulating algorithms [11].

^{*} Corresponding author.

E-mail address: il.im@yonsei.ac.kr (I. Im).

<https://doi.org/10.1016/j.im.2024.104061>

Received 15 August 2023; Received in revised form 29 September 2024; Accepted 12 November 2024

Available online 14 November 2024

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Steering behavior refers to providing feedback to the optimization process of an interactive intelligent system that differs from the user's true preferences to achieve a desired outcome, implying that almost all steering behaviors deviate from true feedback [3]. In contrast, AI manipulation behavior focuses on the intent to manipulate the AI, acknowledging that such behaviors can often align with genuine feedback.

We need a better understanding of AI manipulation behaviors to determine whether AI manipulation benefits or damages firms and individuals. On the one hand, AI manipulation behavior should be encouraged if it positively affects firms and individuals. However, if AI manipulation behavior has negative impacts, we must understand the antecedents that encourage manipulation in order to develop effective countermeasures. Individuals prefer steering AI in their desired direction [12], as evidenced by various articles describing instances of AI manipulation (e.g., [5,6]). Furthermore, users who are highly confident that they can manipulate AI tend to perform better and to be less AI-averse [13]. Explainable AI research has developed a counterfactual explanation technique that provides examples of AI manipulation to increase user trust and performance [14,15]. These examples show the potential positive outcomes of AI manipulability. However, malicious users may intentionally abuse AI manipulation to degrade its performance or manipulate content rankings [11,16,17], emphasizing the need for countermeasures. Moreover, economics researchers have developed algorithms that resist manipulations, based on concerns that strategic manipulations of classification algorithms will damage firms and algorithms [4,18]. However, context has determined whether manipulations have positive or negative impacts on firms, individuals, and algorithms [8,19]. Although previous studies have highlighted the importance of understanding AI manipulation behaviors, we still have limited knowledge about the antecedents of such behaviors. Consequently, we empirically investigate the antecedents of AI manipulation behaviors and explore behavioral differences among users with varying knowledge levels.

Considering that users frequently manipulate AI algorithms to enhance recommendation quality [1,3], we narrow our research to focus on AI recommendation algorithms in popular YouTube and Instagram social media platforms, which were used by more than 4.5 billion people in 2022 [20,21]. We address two research questions regarding AI recommendation algorithms:

RQ1. What antecedents encourage intentions to manipulate AI?

RQ2. Do user knowledge levels affect AI manipulation intentions?

To develop our research, we extensively reviewed the AI manipulation-related literature and based the model primarily on the persuasion knowledge model (PKM), which explains how users perceive and react to AI recommendations [22,23]. We collected survey data from 119 YouTube and 114 Instagram users and analyzed the data using a partial least squares structural equation modeling analysis (PLS-SEM). We found that persuasion knowledge about AI was positively associated with AI manipulation intentions and perceived interactivity with AI. Furthermore, perceived interactivity was positively associated with perceived benefits and negatively associated with costs of AI manipulation, with significant impacts on AI manipulation intention. We also performed a multivariate analysis of variance (MANOVA) and found significant variations in intentions to use different types of AI manipulation behaviors across users with different levels of persuasion knowledge.

Our work has significant implications. As far as we can ascertain, we are the first to empirically explore the antecedents of AI manipulation intention. We find that persuasion knowledge about AI and perceived interactivity with AI significantly shape AI manipulation intentions. The findings have practical implications for platform managers who want to stimulate or discourage AI manipulations.

2. Literature review

Although interest in AI-human interaction is growing, we take a novel approach to exploring AI manipulation. To identify research gaps, we first performed a systematic literature review based on guidelines suggested by Brocke et al. [24] and Webster and Watson [25]. We also investigated (1) ethical AI and misuse behavior to broaden understandings about potential ethical concerns and factors that promote AI mistreatment; (2) adaptation behavior to understand how and why individuals change system usage and manipulation behaviors; and (3) the PKM to understand how knowledge about AI mechanisms determines how users interact with AI interventions such as AI recommendations.

2.1. Systematic literature review on AI manipulation

In the following subsections, we outline the search strategy for literature related to AI manipulation, detailing the search terms, databases, and criteria for inclusion and exclusion.

2.1.1. Search strategy

We focused on journals, reviews, and conference papers written in English and published between 2003 and 2023 to address RQ1 ("What antecedents encourage intentions to manipulate AI?"). We searched for papers in three databases commonly used for information systems literature reviews: Scopus, the Web of Science, and the Association for Information Systems eLibrary (e.g., [26]).

To ensure a comprehensive search on AI manipulation and identify search keywords, we consulted with three experts in AI and online user behavior. Following Ghobadi [27], we used snowball sampling, a method that includes locating relevant papers and examining their reference lists to discover additional keywords used in prior studies (e.g., [28,29]). We constructed the following search strings:

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("ai manipulation*" OR "artificial intelligence manipulation*" OR "algorithm manipulation*" OR "system manipulation*" OR ("strategic manipulation*" OR "strategic classification*" OR "strategic user*" OR "strategic individual*" OR "feature manipulation*" OR "algorithmic recourse*" OR "counterfactual explanation*" OR "consequential decision*") AND ("artificial intelligence" OR "recommender*" OR "system*" OR "algorithm*" OR "machine learning" OR "deep learning") AND ("control*" OR "interact*" OR "manipulat*"))
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2.1.2. Study selection

We used the query to explore titles, abstracts, and keywords. The initial search yielded 938 articles: 596 in Scopus, 338 in the Web of Science databases, and four in the Association for Information Systems eLibrary. After examining titles and abstracts, we identified 99 relevant articles. After removing duplicate and incomplete articles, we had 70 articles. We carefully reviewed their full texts to determine whether their main objectives were to examine or discuss concepts related to AI manipulation. Forty-six articles met the criteria.

2.1.3. Results of the literature analysis

The systematic literature review revealed that AI manipulation studies fall into four categories: strategic classification, counterfactual explanation, adversarial attacks, and others.

Strategic classification research takes an economic perspective to explain that individuals strategically manipulate classification algorithms to achieve desired results based on their understandings of how algorithm inputs and components impact outcomes [4], and on their perceptions that benefits outweigh costs [8,19]. Most researchers assume that strategic manipulation decreases algorithm performance and negatively influences firms and individuals and thus seek robust algorithms that limit such behavior [4,7,18,28,30–41]. However, studies using mathematical models have found that strategic manipulation

behavior can increase classification performance and benefit firms and individuals in some contexts [8,19]. In sum, strategic classification research has focused on the countermeasures and outcomes of strategic manipulation behavior without exploring antecedents of strategic manipulation behavior in the AI context.

Counterfactual explanations are an explainable AI technique. They show users how they might have altered outcomes. For example, “You were denied a loan because your annual income was £30,000. If your income had been £45,000, you would have been offered a loan” ([9], p. 884). Research on counterfactual explanations has mainly focused on how users respond to various types of counterfactual explanations [42–51]. Some studies have compared counterfactual explanations with other explainable AI techniques [14,15,52–62]. Slack et al. [63] showed that counterfactual explanations are vulnerable to manipulations. Taken together, counterfactual explanation research mainly focuses on explaining AI outcomes and on increasing transparency, trust in the system [14], and prediction accuracy [15] by informing users how to strategically manipulate the system.

Two studies showed that AI model security is threatened when adversaries provide misleading input that causes AI models to generate erroneous outputs. Both studies focused on the process of strategic AI manipulation behaviors. Specifically, Khakurel and Rawat [17] showed that AI models are vulnerable to adversarial attacks and Liu et al. [10] developed an adversarial detection framework.

The remaining studies pertain to AI manipulation but fall outside strategic classification, counterfactual explanations, or adversarial attacks categories. Colella et al. [3] examined whether users who better understand interactive intelligent system mechanisms are better at actively providing deceptive responses that control AI output. Shepherd [11] took a rhetorical perspective to examine algorithmic rhetoric in which *Reddit* users strategically manipulate the sorting algorithm to amplify their arguments. Those studies show that people use strategic manipulation to improve their performance in both real-world and experimental settings.

In sum, although the literature offers valuable insights into diverse fields that have actively studied AI manipulation behavior (Table 1), a notable gap remains regarding the antecedents of such behavior.

2.2. Ethical AI and misuse behavior

AI technology advances introduce a plethora of ethical considerations. Fairness, accountability, and transparency (FAT) emerge as particularly essential for shaping trust and satisfaction in AI [64–67]. Along with explainability, FAT positively affects self-efficacy about interacting with AI [66].

AI manipulation intersects with the ethical dimensions of AI usage. Notably, strategic classification research reveals that AI transparency can inadvertently foster strategic manipulation behaviors [8,32], thereby compromising AI fairness and necessitating corporate accountability [28,41]. Explainable AI studies have also delved into

methods to enhance AI model explainability and transparency [42,57].

From an AI ethics perspective, AI manipulation is a form of intentional information technology (IT) misuse, irrespective of explicit malicious intent [68–70]. Acts such as using unlicensed software and gaining unauthorized access have been traditionally considered IT misuse, but strategic manipulators intentionally steer AI toward specific outcomes for personal gain or advantage. In both cases, technology is deliberately leveraged in a manner that diverges from ethical guidelines or intended purposes. Previous research has demonstrated that IT misuse behaviors are motivated by psychological factors such as perceived behavioral control, perceived overall gain, subjective norms, and attitudes [69,71].

Taken together, this diverse literature shows that AI manipulation is profoundly related to ethical AI and aligned with IT misuse. However, the research has mostly focused on the ethical dimensions of AI or the determinants of IT misuse, while overlooking the need to examine factors that affect AI manipulation.

2.3. Adaptation behavior

The adaptation behavior literature has shown that as individuals gain system experience and knowledge, they use systems more efficiently; thus, feedback loops occur between systems and users [72,73]. Adaptation is a coping behavior in which individuals strive to handle specific outcomes arising from IT-related events [74]. Perceived IT innovativeness, self-efficacy, usefulness, and facilitating conditions have been positively associated with adaptation intentions [73,75–77]. Furthermore, adaptation has been shown to positively affect job performance [75].

Adaptation behavior has been classified into several types, such as extended, integrative, and innovative use (i.e., emergent use) [78]. Innovative adaptation behavior is most closely related to our research because it occurs when individuals use system features in innovative ways that differ from the original purposes [73,79]. Users who have IT knowledge tend to recognize the value of IT and to assimilate it in ways that lead to innovative IT use and performance [79]. Some researchers have investigated innovative use behavior in the AI context. For example, Van de Wetering [80] found that innovative AI use is positively associated with organizational strategic flexibility, and Wang et al. [81] found it to be related to internal and external AI agility. Wang et al. [82] showed that trust in AI significantly influences innovative AI use.

Amershi et al. [1] explained how individuals adaptively and strategically use interactive machine learning algorithms in some cases. They emphasized that the algorithm’s learning cycles should be designed to provide rapid and immediate responses to input data, to focus on a certain aspect of the model, to make incremental changes without drastic alterations, and to be used adaptively. The findings emphasize the need for responsiveness and personalization.

Adaptation behaviors have been studied in various contexts such as corporate IT systems [74,75], Microsoft Office [73], and social media [76]. However, few studies have delved into adaptation behavior or innovative use behavior in the AI context. In addition, we still lack understanding of the antecedents of adaptation behavior in AI contexts. Furthermore, to reiterate, knowledge of and interactivity with the system are major factors that influence adaptation behavior [1,72,73], but we lack research regarding how these factors relate to AI manipulation behavior.

2.4. The PKM

The PKM is often used to explain how consumers, as persuasion targets, judge whether advertisers, as persuasion agents, have ulterior motives in persuading consumers to adopt “beliefs, decisions, or actions” ([22], p. 2). The model is also used to explain how consumers cope with persuasion attempts: they can choose the most appropriate coping behaviors if they know about persuasive methods [22]. In the context of AI

Table 1
Systematic literature review: a summary.

Research stream	Topic	References
Economic studies: Strategic classification	Developing robust algorithms to counter strategic behaviors Strategic manipulation effects on firms, users, and algorithms	[4,7,18,28, 30–41] [8,19]
Explainable AI studies: Counterfactual explanation	Developing counterfactual explanations Comparing counterfactual explanations with other explainable AI techniques Vulnerabilities of counterfactual explanations	[42–51] [14,15, 52–62] [63]
Security studies: Adversarial attacks		[10,17]
Other studies: Steering behavior, algorithmic rhetoric		[3,11]

recommendations, users gradually gain knowledge about how AI learns their preferences and suggests recommendations through continual interactions with AI recommendation algorithms. Here, AI recommendations are a persuasion method for prompting users to click on content (e.g., YouTube videos), to peruse ads, or to purchase products [23,83]. Users who understand AI may manipulate it as a coping mechanism against AI recommendations.

Persuasion knowledge (PK) means that persuasion targets understand persuasion attempts [84]. PK is a multidimensional concept that includes awareness and understanding about persuasion, perceptions of the effectiveness and appropriateness of persuasion, and coping strategies for countering persuasion attempts [22,83]. Individuals generally acquire PK by directly experiencing, learning, or observing persuasion strategies [22,23].

Although PKM is traditionally used to study advertising (e.g., [85–89]), it has also been applied to understand user behaviors in diverse contexts such as fake news [90], pricing strategies [91], online reviews [92], health-related websites [93], compliance with information security policies [94], and recommendation systems [23].

In the context of AI advertising (online behavioral advertising), PK was shown to positively influence self-efficacy, third-person perception, benefits from personally relevant information, privacy risks and concerns, skepticism, and coping behaviors such as advertisement avoidance [83,84,95]. In the AI recommendation context, PK was shown to generate positive behavioral responses to recommendations for using the system and perceptions of persuasion appropriateness [23]. Most PKM studies focusing on AI have adopted the benefit and risk assessment perspective to explain why PK influences coping behaviors [23,83,84,95]. Moreover, PKM research in non-AI contexts has shown that PK is significantly associated with interactivity, diagnosticity, affective resistance, authenticity, trust, attitude, and behavioral intention [85,86,89,90,92,93,96]. In sum, PK correlates highly with the benefits and risks of persuasion attempts and with self-efficacy, personalization, interactivity, privacy concerns, trust, attitude, and behavioral intention.

Tangwaragorn [23] showed that users who have PK are more likely to accept AI-recommended products. However, the research was designed for purposes other than capturing AI manipulation behaviors or for investigating whether PK encourages AI manipulation. Hence, even though much PKM research has been conducted in the AI context, we are the first, to the best of our knowledge, to apply PKM to AI manipulation.

The literature review indicated that PK comprises *awareness* or *understanding*. If we focus on *awareness*, PK includes attitudes toward and recognition of persuasion attempts [23,85–87,89,90,92,96,97]. If we focus on *understanding*, PK comprises cognitive insight into persuasion attempts [83,84,88,93,95,98]. Interestingly, PKM research in the

context of AI recommendation systems has overlooked the *understanding* definition of PK (e.g., [23]).

3. Hypothesis development

In this study, we investigate AI manipulation behavior primarily based on the PKM and on findings from misuse and adaption behavior research. The PKM explains how user knowledge regarding the AI mechanism drives users to adopt coping behavior such as AI manipulation in response to AI persuasion attempts (e.g., AI recommendations) [23,99]. Our literature review indicated that PK and perceived interactivity are major antecedents in predicting AI manipulation. Fig. 1 shows our research model.

3.1. Persuasion knowledge effects on perceived interactivity

AI is an interactive decision aid [100,101] and a persuasion system [23,99]. In the context of AI recommendation algorithms, interactivity is a key factor determining whether users react positively [102], while personalization and responsiveness are significant factors affecting AI manipulation behavior [1]. In addition, knowledge of and interactivity with the system significantly affect AI manipulation and adaptation intention [3,11,33,41,72,73].

Individuals who have PK believe that they understand how AI generates recommendations. Subjective PK was shown to be significantly and positively associated with objective PK [84,98]. We believe that individuals who have PK will also have *perceived interactivity* in believing that they can efficiently interact with AI [103–105]. Perceived interactivity has three dimensions: perceived personalization, perceived control, and perceived responsiveness (a second-order construct) [105–107]. *Perceived personalization* means that individuals believe AI is appropriately customized for their use [23,102,105,106]. *Perceived control* means that individual believe they command over AI [103,106], which is also referred to as self-efficacy [108]. *Perceived responsiveness* means that individuals believe that AI responds to their inputs [105,106].

Individuals who have significant PK perceive AI as effective and personalized [22,83]. They are confident that they can control AI [83] and that AI responds to their input behavior quickly and appropriately. Thus, they feel that AI is personalized, controllable, and appropriately responsive.

Empirical research supports our contention that individuals who have higher PK regarding AI have higher perceptions about interactivity with the algorithms. For example, PK has been shown to positively affect individual expectations of receiving relevant and personalized information [23,83,84,95]. Moreover, PK is positively associated with

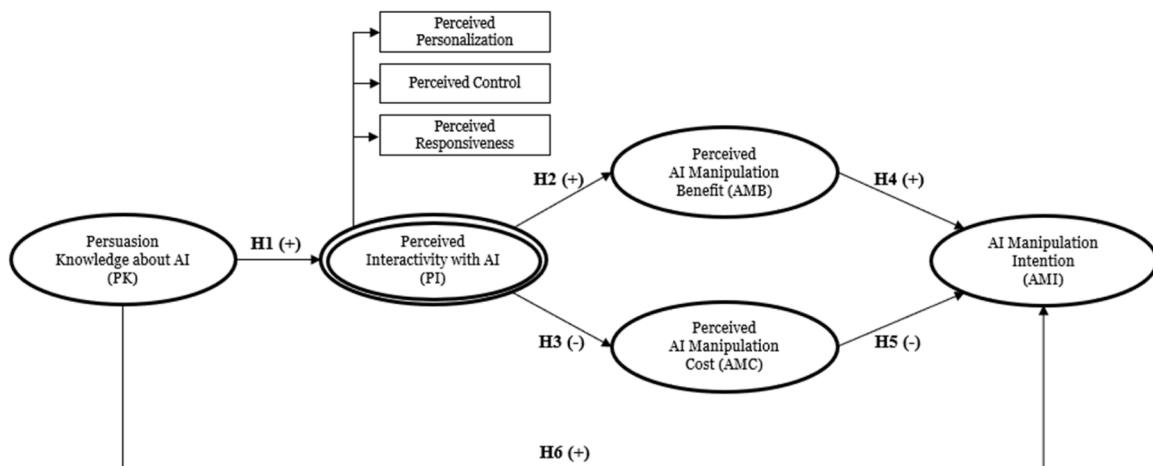


Fig. 1. Conceptual model.

perceived control [83,90] and perceived interactivity [96,97]. Thus, users who have PK recognize that the algorithms provide interactivity, personalization, control, and responsiveness, prompting our first hypothesis:

H1. *Persuasion knowledge about AI is positively associated with perceived interactivity with AI.*

3.2. Perceived interactivity effects on perceived benefits and costs of AI manipulation

To manipulate AI and exploit AI's interactive nature, users must actively modify their inputs after observing results [4,41]. However, rational users can assess whether efforts to manipulate AI equal the benefits [8,30,33,34]. Hence, perceived AI manipulation costs and benefits determine whether perceived interactivity relates to AI manipulation intention. We contend that the operational definition of *perceived AI manipulation benefit* indicates whether individuals perceive that AI manipulation will be beneficial. *Perceived AI manipulation cost* indicates expectations regarding how much effort they must invest.

We believe that perceived interactivity is significantly associated with perceptions of benefits and costs of AI manipulation because individuals who perceive AI as being more interactive (i.e., personalized, controllable, and responsive) are aware that AI is useful and view AI manipulation as an attractive, beneficial, and easy means to improve AI accuracy, worthy of investing effort. AI responsiveness and personalization are important factors in strategic adaptation behavior [1]. In addition, perceived interactivity is positively associated with perceived benefits such as performance expectancy, usefulness, effectiveness, and usability [103,109–111]. Perceived personalization positively impacts perceived assistance [23]. When AI was studied in the context of a recommendation agent, perceived responsiveness was positively associated with perceived outcome expectancy (benefits) and negatively associated with the loss of privacy (costs) [112]. Hence, we formulated H2 and H3:

H2. *Perceived interactivity with AI is positively associated with perceived benefits of AI manipulation.*

H3. *Perceived interactivity with AI is negatively associated with perceived costs of AI manipulation.*

3.3. AI manipulation intentions in response to perceived benefits and costs of AI manipulation

Decisions to undertake strategic manipulation behaviors are based on assessments of the associated costs and benefits [8,30,33,34]. Several PKM studies have adopted cost-benefit framework perspectives [94] and risk-benefit frameworks in their models [23,83,84,95]. The adaptation and misuse behavior literature has shown that perceived benefits such as perceived overall gain, opportunity, and usefulness are positively related to behavioral intentions [69,75–77]. In addition, Beaudry and Pinsonneault [74] posited that individuals used adaptation behavior as a coping strategy to deal with an IT event after they appraised expected outcomes. Therefore, we expect that perceived AI manipulation benefits and costs significantly influence AI manipulation intention, indicating how extensively individuals intend to continue manipulating AI in the future, leading us to hypothesize:

H4. *Perceived benefit of AI manipulation is positively associated with intentions to manipulate AI.*

H5. *Perceived cost of AI manipulation is negatively associated with intentions to manipulate AI.*

3.4. Persuasion knowledge effects on AI manipulation intention

Individuals who have strong knowledge about AI mechanisms are more likely to adapt their input behavior to receive desirable outcomes [41]. Transparency increases knowledge about AI and leads to more strategic manipulation behaviors [8,32]. In addition, users who know how algorithms sort content were shown to leverage their knowledge to push content to the top [11]. PK has also been significantly associated with intentions to act. For example, Manika et al. [93] found that PK regarding health advice/guidelines positively affected intentions to control weight. Thus, we postulate:

H6. *Persuasion knowledge about AI is positively associated with intentions to manipulate AI.*

4. Methodology

In this study, we investigated the factors that affect intentions to manipulate AI. Based on recent online articles explaining how users can manipulate AI on YouTube and Instagram (e.g., [5,6]), and considering that various age groups widely use those platforms [113,114], we surveyed YouTube and Instagram platform users 14 to 59 years old.

4.1. AI manipulation examples on YouTube and Instagram

In this section, we show examples of AI manipulation appearing on YouTube and Instagram platforms, where users manipulate AI in various ways. For example, they may deliberately click on platform-recommended video content that does not actually interest them but falls within a genre that they prefer so that they will receive similar recommendations. The same motivations may cause them to indicate that they “like” recommended videos or to follow certain content creators. Conversely, they may intentionally “dislike” recommended video content to avoid recommendations for similar content. Fumba [5] and Gewlertz [6] detail further instances of AI manipulation on YouTube and Instagram platforms.

4.2. Survey procedure

Our survey was designed to minimize the potential bias that would occur if we combined data from YouTube and Instagram platforms. Thus, in the first stage, we asked respondents to indicate which platforms they currently use. Those who use platforms other than YouTube or Instagram were told to exit the survey. The remaining participants were asked which platform they used most recently. Survey participants viewed an example of an AI recommendation algorithm and a description of AI manipulation. They completed measurements for the demographic, control, independent, mediator, and dependent variables, and three attention checks spaced throughout the survey (Fig. 2).

YouTube and Instagram have many types of pages. To ensure that all survey respondents considered the same type of page, we explicitly told respondents which pages resulted from AI recommendation algorithms by providing an example of pages featuring recommendation algorithms. For example, respondents learned that YouTube applies the AI recommendation algorithm to its content page and Instagram applies it to its explore tab (Figs. 3 and 4).

We also told survey respondents about AI manipulation by providing the AI manipulation description (Fig. 5). However, the AI manipulation description may influence responses to PK, interactivity, and the control variables. Therefore, they answered the measurements of these variables first. Then we provided the AI manipulation description and asked participants to answer questions about their intentions and their perceived AI manipulation benefits and costs.

To determine whether respondents correctly read and understood the AI recommendation algorithm and AI manipulation descriptions, we asked three attention checks spaced throughout the survey: (1) “Does

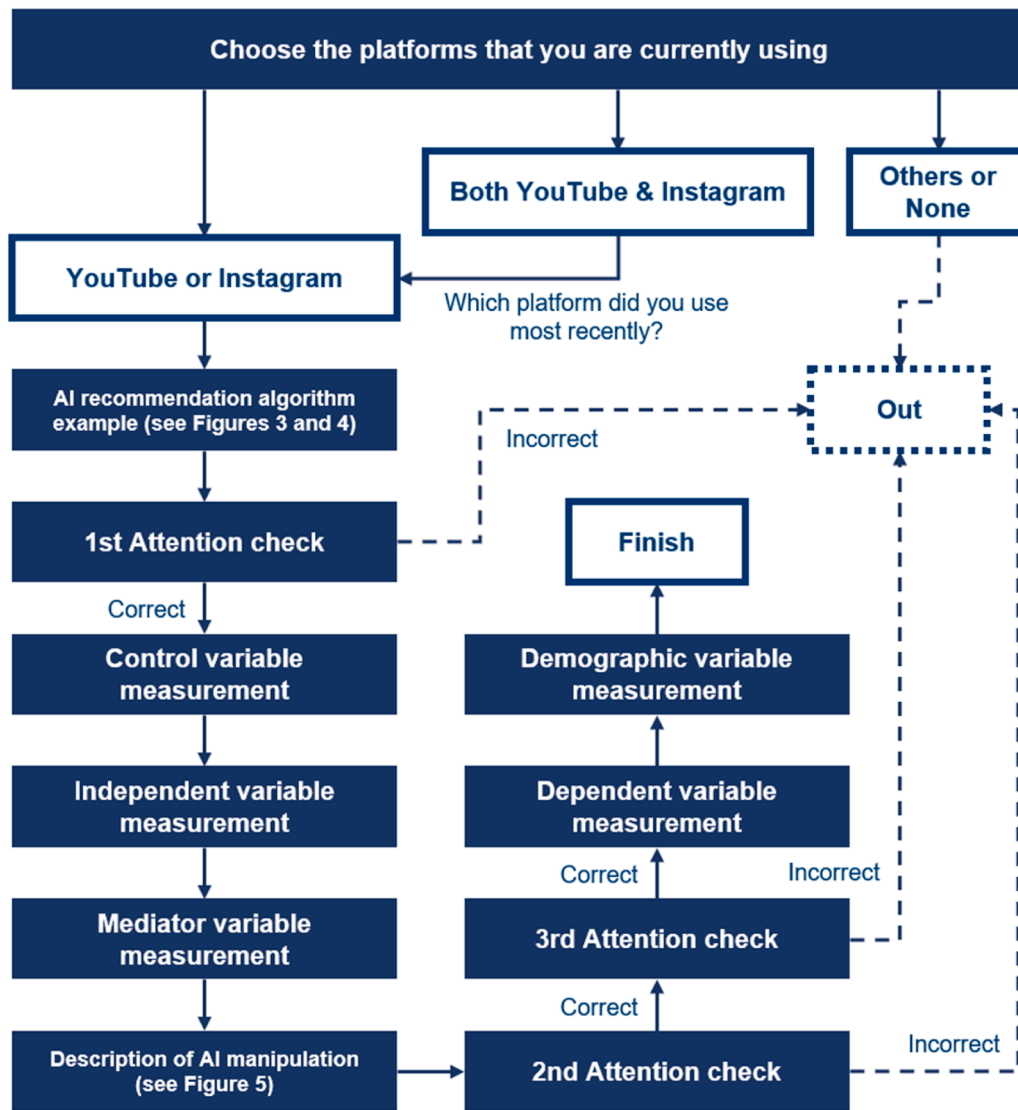


Fig. 2. Survey procedure.

YouTube (Instagram) use an AI recommendation algorithm that recommends videos through the recommended content list?" (2) "Which of the types shown in Fig. 5 do not include direct manipulations of the AI recommendation algorithms?" (3) "Which platform is this survey asking about?" Among the 364 initial respondents, 29 respondents failed the first attention check; 82 failed the second; and 20 failed the third. They were eliminated, leaving 233 valid respondents.

4.3. Instrument development

All items related to perceptions were measured on a seven-point Likert scale from 1 (strongly disagree) to 7 (strongly agree). The measurement items are listed in Table 2. The PK items were slightly adapted from Bearden et al. [115], Ham [83], and Ryu and Park [98]; the perceived personalization (PP) items were slightly adapted from Ham [83] and Tangwaragorn [23]; the perceived responsiveness (PR) and perceived behavioral control (PC) items were slightly adapted from Lee et al. [103] and Song and Zinkhan [106]; the AI manipulation benefit (AMB) items were slightly adapted from Bulgurcu et al. [94]; the two (AMC1 and AMC2) variants out of four AI manipulation benefit (AMC) items were slightly adapted from Bulgurcu et al. [94]. We developed the remaining AI manipulation benefit items for this research. The AI manipulation intention (AMI) items were slightly adapted from

Bhattacharjee [116] and Venkatesh et al. [117]; the content involvement (INV) items were slightly adapted from Kim et al. [118]; the platform loyalty (LOY) items were slightly adapted from Flavián et al. [119]; and the privacy concern (PVC) item was slightly adapted from Ham [83]. We also measured age, gender, education, internet usage, platform experience, and platform type as control variables. We developed new survey questions about intentions to "click" or press the "like," "dislike," and "follow" buttons on the content (CLICK, LIKE, DISLIKE, and FOLLOW).

All items were originally written in English. After items were compiled into the survey questionnaire, they were translated into Korean. Back-translation determined that the English-to-Korean translation was reliable. An extensively experienced translator back-translated the Korean instruments into English. The author and the translator discussed and corrected differences between the back-translated and original questionnaires to ensure that the Korean translation was accurate and appropriate.

Next, we surveyed 12 YouTube or Instagram users to evaluate the clarity of the descriptions regarding the AI recommendation algorithm examples and AI manipulation. Both types of users equally indicated that the descriptions were clear and easily understandable. We also conducted a pilot study with 24 YouTube or Instagram users who were MS or PhD students. Their comments led us to modify several items to

YouTube uses an AI recommendation algorithm to provide a recommended video list. When users access content with a computer, the list appears to the right of videos. When they access content with a mobile device (right), the list appears below videos. Scrolling down the list shows more recommended content.

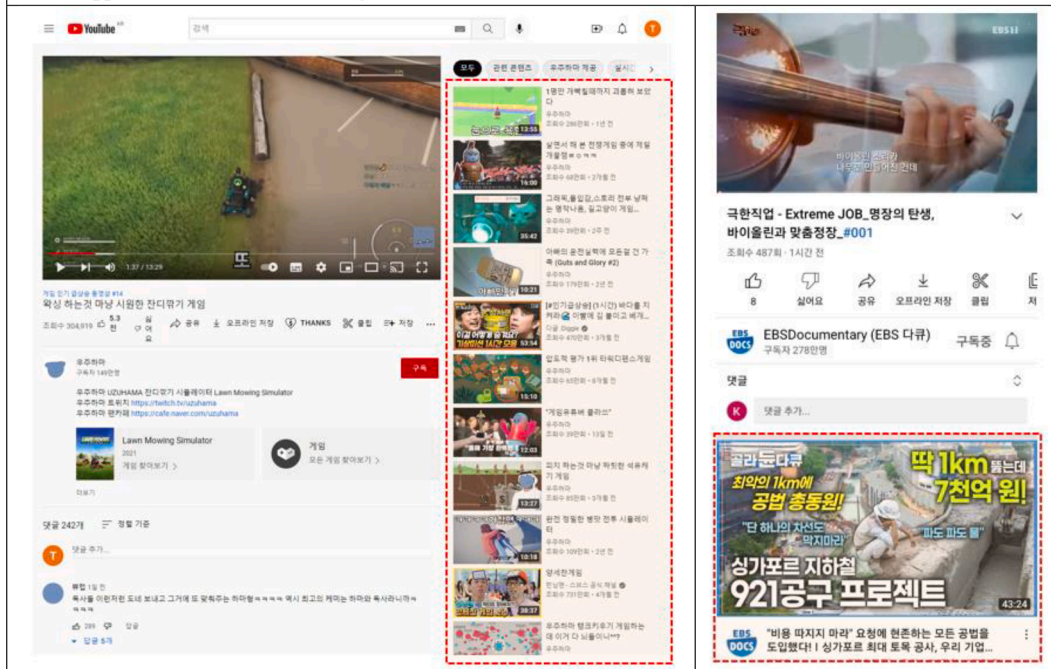


Fig. 3. Example of an AI recommendation algorithm on YouTube.

Instagram uses an AI recommendation algorithm to provide a recommended post list, accessed through the Explore tab. When users access content with a computer, they access the list by clicking on the compass icon. When they access content with a mobile device, they access the list by using the magnifying glass icon. Scrolling down the list shows more recommended content.

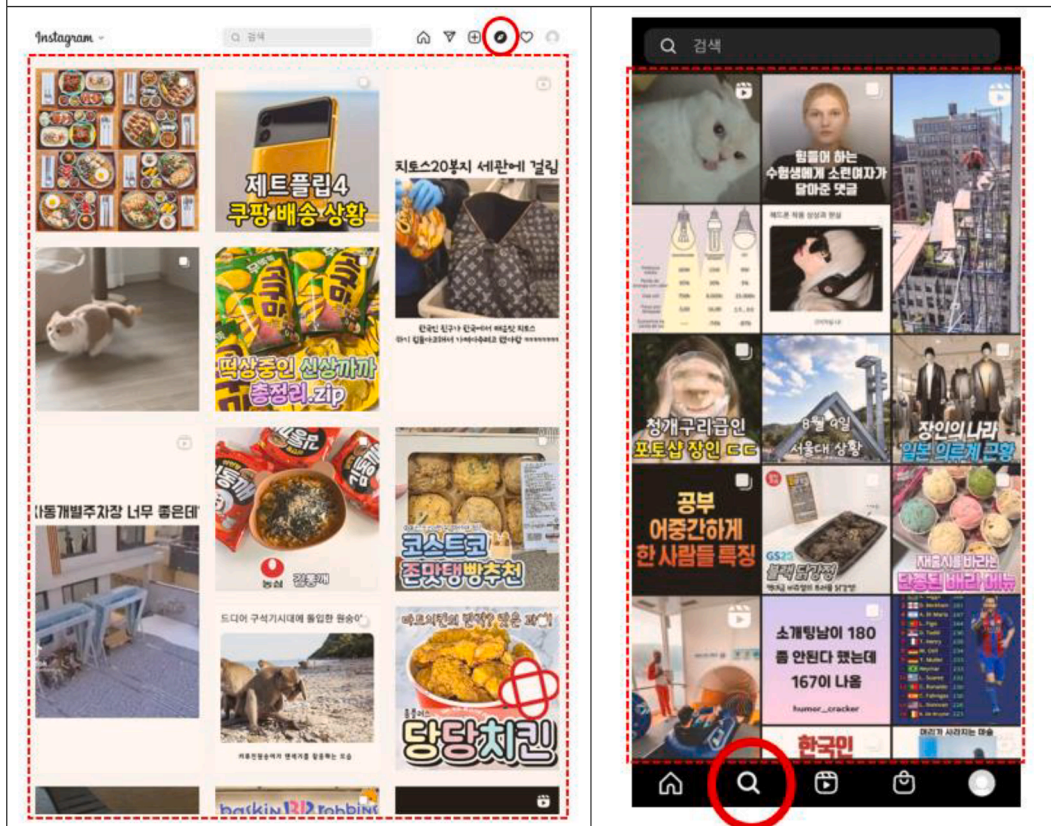


Fig. 4. Example of an AI recommendation algorithm on Instagram.

Users may directly and intentionally manipulate AI recommendation algorithms to recommend certain types of content (rather than evaluating the content) by:

Example 1) Intentionally pressing the “Like,” “Dislike,” “Save to favorites list,” “Not interested,” and “Share with others” buttons for specific content.

Example 2) Intentionally leaving a comment for selected content.

Example 3) Intentionally pressing the “Subscribe” or “Follow” buttons for a specific content uploader.

Example 4) Intentionally clicking on recommended content.

Fig. 5. Description of AI manipulation.

improve clarity and to delete several items that lacked relevance.

4.4. Data collection

In September 2022, we collected data from users of YouTube or Instagram through a Korean online market research company that features 1.6 million online panels. After we dropped 131 for failing the attention checks as mentioned in Section 4.2, the final dataset included 233 valid response sets: 119 YouTube users and 114 Instagram users. Table 3 shows descriptive statistics regarding respondent characteristics.

We conducted two-sample *t*-tests to identify potential differences between YouTube and Instagram users on all variables used in this study and found no significant differences in the main variables: persuasion knowledge, perceived personalization, perceived responsiveness, perceived behavioral control, and perceived AI manipulation benefit, cost, and intention. The findings suggest that platform type had no large effects on our main findings. However, we found significant differences in platform experience ($t = 3.708$), platform loyalty ($t = 3.314$), content involvement ($t = 5.282$), and FOLLOW ($t = 3.158$). YouTube users had higher perceptions of platform experience and platform loyalty, perhaps because YouTube has a longer history and a larger user base [21]. They may have indicated higher content involvement and intention to FOLLOW because YouTube focuses on longer and more general video content, which reduces privacy concerns. In contrast, Instagram provides personal photos and shorter videos. The focus on personal life content makes it harder to “follow” other users.

5. Data analysis and results

5.1. Measurement validation

To test the second-order constructs, we used PLS-SEM in SmartPLS v.4.0.9.4 [120]. Scale reliability, internal consistency, and convergent validity were determined by calculating the factor loadings, Cronbach’s alpha, composite reliability, and average variance extracted (AVE), respectively. The standardized path loadings of all constructs were greater than the suggested cutoff value of 0.7 [121], except for PP5 (0.671) and PR5 (0.693), which were still close to the cutoff (Table 2). The composite reliability values and Cronbach’s alpha values for each construct were greater than the cutoff value of 0.7. All AVEs were greater than 0.5, indicating acceptability [121] (Table 4). The results indicated satisfactory data convergent validity. Discriminant validity was assessed using both Fornell and Larcker’s criteria [122] and the heterotrait-monotrait (HTMT) criteria [123]. The square roots of the AVEs were greater than the inter-construct correlations, indicating that we met Fornell and Larcker’s criteria. In addition, all HTMT values were less than the threshold of 0.85, suggesting that the data had discriminant validity (Table 5). Multicollinearity was no serious concern, given that the highest variable inflation factor value (2.401) was less than the recommended cutoff value of three [124]. Finally, Harman’s single-factor test was used to test common method bias [125]. The

largest single factor of the total variance (27.033 %) was less than the cutoff criterion of 50 %, indicating no significant common method bias. Following MacKenzie et al. [126], the second-order construct (i.e., perceived interactivity) was validated. Its AVE, composite reliability, and Cronbach’s alpha were 0.630, 0.836, and 0.712, respectively. The factor loadings of perceived interactivity were statistically significant ($p < 0.001$) and greater than the cutoff value of 0.7. Those test results indicated that the second-order model was reliable and valid.

5.2. Hypotheses testing

The hypotheses were tested using two-tailed *t*-tests with a bootstrap procedure with 5,000 resamples. Fig. 6 shows structural equation modeling results. PK was positively associated with perceived interactivity ($\beta = 0.505$, $p < 0.001$), supporting H1: *Persuasion knowledge about AI is positively associated with perceived interactivity with AI*.

Perceived interactivity was positively associated with perceived AI manipulation benefits ($\beta = 0.251$, $p < 0.01$) and negatively associated with perceived costs ($\beta = -0.228$, $p < 0.01$), supporting H2: *Perceived interactivity with AI is positively associated with perceived benefits of AI manipulation* and H3: *Perceived interactivity with AI is negatively associated with perceived costs of AI manipulation*.

Perceived AI manipulation benefit positively affected AI manipulation intention ($\beta = 0.387$, $p < 0.001$), while cost ($\beta = -0.245$, $p < 0.001$) negatively affected AI manipulation intention, supporting H4: *Perceived benefit of AI manipulation is positively associated with intentions to manipulate AI* and H5: *Perceived cost of AI manipulation is negatively associated with intentions to manipulate AI*.

PK regarding AI significantly and positively affected AI manipulation intentions ($\beta = 0.240$, $p < 0.01$), supporting H6: *Persuasion knowledge about AI is positively associated with intentions to manipulate AI*.

Regarding the control variables, platform loyalty was positively associated with AI manipulation intentions ($\beta = 0.2$, $p < 0.05$). Given that platform loyalty is positively related to platform continuance usage intention [127], platform loyalty may have a positive impact on AI manipulation intention through platform continuance usage intention. That is, individuals who continue to use the platform are more likely to manipulate the platform’s AI because the benefits of AI manipulation would be greater for those loyal users who interact with AI more often than others. The other control variables were nonsignificant. Our model explains 25.6 %, 6.3 %, 5.2 %, and 41.9 % of the variance (R^2) in the perceived interactivity, perceived AI manipulation benefit, perceived AI manipulation cost, and AI manipulation intention, respectively.

5.3. Mediation analysis

We performed an additional analysis to examine whether perceived interactivity and perceived AI manipulation benefits and costs mediate PK’s effect on manipulation intentions. A two-step procedure using bootstrapping tested mediating effects of the variables [128]. Table 6 provides a summary: the indirect paths are significant at the significance level of $p < 0.05$. For example, PK positively affected AI manipulation

Table 2
Construct reliability and validity.

Item code	Item	Loading
PK1	I have no trouble understanding how YouTube's (or Instagram's) recommendation algorithm works.	0.786
PK2	I know how YouTube's (or Instagram's) recommendation algorithm offers tailored content to me using behavior tracking on the platform.	0.858
PK3	I understand how YouTube's (or Instagram's) recommendation algorithm shows tailored content to me using behavior tracking on the platform.	0.887
PK4	I can see through the mechanism of YouTube's (or Instagram's) recommendation algorithm used to suggest me some content.	0.905
PK5	Compared to others, I am knowledgeable about how YouTube's (or Instagram's) recommendation algorithm works.	0.842
PP1	I feel that YouTube's (or Instagram's) recommendation algorithm enables me to know content that I am interested in.	0.863
PP2	Overall, YouTube's (or Instagram's) recommendation algorithm is tailored to my situation.	0.888
PP3	YouTube's (or Instagram's) recommendation algorithm is customized to my needs.	0.891
PP4	YouTube's (or Instagram's) recommendation algorithm makes recommendations that match my needs.	0.916
PP5	YouTube's (or Instagram's) recommendation algorithm does not know what I want. (R)	0.671
PC1	I know how to control YouTube's (or Instagram's) recommendation algorithm efficiently.	0.866
PC2	Overall, I can control YouTube's (or Instagram's) recommendation algorithm well.	0.924
PC3	YouTube's (or Instagram's) recommendation algorithm is manageable.	0.896
PC4	While I am using YouTube's (or Instagram's) recommendation algorithm, I can freely choose what I want to see.	0.775
PR1	YouTube's (or Instagram's) recommendation algorithm quickly responds to my input, such as (e.g., clicking).	0.779
PR2	YouTube's (or Instagram's) recommendation algorithm gives relevant information with respect to my input (e.g., clicking).	0.744
PR3	YouTube's (or Instagram's) recommendation algorithm has no delay in operations (e.g., loading time).	0.825
PR4	YouTube's (or Instagram's) recommendation algorithm processes my inputs (e.g., clicking) very quickly.	0.873
PR5	YouTube's (or Instagram's) recommendation algorithm is very slow in responding to my requests. (R)	0.693
PR6	Getting recommendations from YouTube's (or Instagram's) recommendation algorithm is very fast.	0.848
AMB1	Direct manipulation of YouTube's (or Instagram's) recommendation algorithm would be favorable to me.	0.860
AMB2	Direct manipulation of YouTube's (or Instagram's) recommendation algorithm would be beneficial for me.	0.918
AMB3	Direct manipulation of YouTube's (or Instagram's) recommendation algorithm would create advantages for me.	0.945
AMB4	Direct manipulation of YouTube's (or Instagram's) recommendation algorithm would provide gains to me.	0.915
AMC1	Direct manipulation of YouTube's (or Instagram's) recommendation algorithm is burdensome for me.	0.732
AMC2	Direct manipulation of YouTube's (or Instagram's) recommendation algorithm is time consuming for me.	0.847
AMC3	Direct manipulation of YouTube's (or Instagram's) recommendation algorithm requires a lot of mental effort for me.	0.911
AMC4	I think that direct manipulation of YouTube's (or Instagram's) recommendation algorithm is stressful for me.	0.921
AMI1	I intend to continue directly manipulating YouTube's (or Instagram's) recommendation algorithm in the future.	0.891
AMI2	I will always try to directly manipulate YouTube's (or Instagram's) recommendation algorithm in my daily life.	0.932
AMI3	I plan to continue to directly manipulate YouTube's (or Instagram's) recommendation algorithm frequently.	0.903
AMI4	If I could, I would like to continue directly manipulating YouTube's (or Instagram's) recommendation algorithm.	0.885

Table 2 (continued)

Item code	Item	Loading
INV1	Watching YouTube (or viewing Instagram) content is one of the most enjoyable things I do.	0.938
INV2	Watching YouTube (or viewing Instagram) content is important to me.	0.920
INV3	Watching YouTube (or viewing Instagram) content is pleasurable to me.	0.941
LOY1	I visit YouTube (or Instagram) more frequently than others like me.	0.833
LOY2	YouTube (or Instagram) is the main platform where I access content.	0.944
LOY3	YouTube (or Instagram) is my favorite platform for accessing content.	0.905
PVC	I think companies share my online behavioral data without my permission.	-
CLICK	I think I will click on more content on YouTube's (or Instagram's) recommendation list in the future.	-
DISLIKE	I think I will click the "dislike" button on more YouTube (or Instagram) content in the future.	-
FOLLOW	I think I will subscribe to (or follow) more YouTube (or Instagram) content uploaders in the future.	-
LIKE	I think I will click the "like" button on more YouTube (or Instagram) content in the future.	-

Note: (R) indicates reverse-coded items. PK (persuasion knowledge), PP (perceived personalization), PC (perceived control), PR (perceived responsiveness), AMB (AI manipulation benefit), AMC (AI manipulation cost), AMI (AI manipulation intention), INV (content involvement), LOY (platform loyalty), PVC (privacy concern), CLICK (intention to click on more content), DISLIKE (intention to press the "dislike" button for more content), FOLLOW (intention to subscribe to or follow more content uploaders), LIKE (intention to press the "like" button for more content).

Table 3
Descriptive statistics.

Name	Details	YouTube (n= 119)		Instagram (n= 114)	
		Frequency	%	Frequency	%
Gender	Man	59	49.6	55	48.2
	Woman	60	50.4	59	51.8
Age	10–19 years old	22	18.5	22	19.3
	20–29 years old	23	19.3	24	21.1
	30–39 years old	24	20.2	27	23.7
	40–49 years old	27	22.7	23	20.2
	50–59 years old	23	19.3	18	15.8
Education	High school graduate or less	34	28.6	30	26.3
	2- or 3-year college graduate	16	13.4	16	14.0
	Undergraduate	54	45.4	51	44.7
	Postgraduate	15	12.6	17	14.9
Platform Experience	1–3 months	0	0	3	2.6
	3–6 months	0	0	7	6.1
	6–9 months	4	3.4	6	5.3
	9–12 months	2	1.7	7	6.1
	More than 12 months	113	95.0	91	79.8
Internet Usage (per day)	0–2 hours	7	5.9	6	5.3
	2–4 hours	39	32.8	42	36.8
	4–6 hours	33	27.7	38	33.3
	6–8 hours	18	15.1	14	12.3
	More than 8 hours	22	18.5	14	12.3

intention through perceived interactivity and through perceived AI manipulation benefit or cost. Similarly, PK positively influenced perceptions of AI manipulation benefits and negatively influenced perceptions of costs through perceived interactivity. Perceived AI manipulation benefits and costs had statistically significant indirect effects on how perceived interactivity related to AI manipulation intention. These findings suggest that PK's impact on AI manipulation intentions is partially mediated by perceived interactivity, benefits, and costs.

Table 4
Reliability and validity.

	Cronbach's alpha	CR	AVE
PK (persuasion knowledge)	0.909	0.932	0.734
PP (perceived personalization)	0.903	0.928	0.723
PC (perceived control)	0.888	0.923	0.752
PR (perceived responsiveness)	0.884	0.912	0.634
AMB (AI manipulation benefit)	0.931	0.951	0.829
AMC (AI manipulation cost)	0.878	0.916	0.733
AMI (AI manipulation intention)	0.924	0.946	0.815
INV (content involvement)	0.925	0.953	0.870
LOY (platform loyalty)	0.877	0.924	0.802

Therefore, we infer that PK and perceived interactivity are key factors affecting intentions to manipulate AI and that a cost-benefit framework is a suitable approach for understanding AI manipulation intentions.

5.4. Path comparison tests

Because we adopted the cost-benefit framework, we additionally investigated which paths more significantly affect AI manipulation intention. Specifically, we compared the path coefficients of the manipulation benefit, cost, and PK on AI manipulation intention, following the comparison equation for path coefficients [129,130]. The three variables showed significant differences (Table 7). Perceived benefit had the greatest impact on manipulation intention, followed by PK and perceived cost. These findings suggest that perceived AI manipulation benefit affects AI manipulation intention more than perceived AI manipulation cost.

5.5. Multivariate analysis of variance (MANOVA)

Our second research question was “Do user knowledge levels affect AI manipulation intentions?” To answer the question, we examined how PK affects intentions to engage in CLICK, LIKE, DISLIKE, and FOLLOW AI manipulation behaviors, which are considered representative behaviors in AI recommendation algorithms [5,6]. We conducted a MANOVA in which PK was classified as low ($n=94$), medium ($n=61$), or high ($n=78$). Table 8 summarizes the descriptive statistics. We first checked the MANOVA assumptions. Levene's F -test (Table 8) and Box's M -test (M -statistic= 28.761, $p=0.109$) provided statistically nonsignificant results, indicating that the MANOVA assumptions were satisfied [131].

PK level had statistically significant multivariate effects on AI manipulation intentions (Pillai's trace= 0.120, F -statistics= 3.623, $p<0.001$). The univariate effects were significant for CLICK (F -statistics= 8.725, $p<0.001$), LIKE (F -statistics= 9.070, $p<0.001$), DISLIKE (F -statistics= 3.312, $p<0.05$), and FOLLOW (F -statistics= 9.222, $p<0.001$). As Table 8 shows, post hoc tests compared PK levels for differing manipulation intentions. High and low PK levels showed significant differences for CLICK and DISLIKE intentions only. However, high, medium, and low PK levels showed statistically significant differences for LIKE and FOLLOW intentions. In other words, users who were highly knowledgeable about the algorithm were more likely than less knowledgeable users to intend to practice LIKE and FOLLOW manipulations.

These findings suggest that LIKE and FOLLOW are more advanced AI manipulation techniques than CLICK and DISLIKE. We suspect that knowledgeable users will prefer the most powerful feedback, LIKE and FOLLOW signals, to indicate explicit preferences. Conversely, CLICK is a relatively passive action indicating user interest or engagement with specific content but failing to provide the most explicit and feedback or preference signals. In addition, DISLIKE indicates only disfavor without indicating actual preference.

Table 5
Discriminant validity.

	Fornell and Larcker's criteria									HTMT criteria								
	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
1	0.867																	
2	0.116	0.91									0.127							
3	-0.221	-0.055	0.856								0.24	0.071						
4	0.297	0.091	-0.132	0.933							0.331	0.096	0.14					
5	0.275	0.112	-0.161	0.732	0.895						0.307	0.122	0.18	0.803				
6	0.44	0.258	-0.21	0.447	0.405	0.85					0.486	0.26	0.233	0.428	0.464			
7	0.513	0.182	-0.14	0.192	0.249	0.375	0.857				0.562	0.198	0.141	0.206	0.275	0.397		
8	0.33	0.249	-0.082	0.384	0.377	0.584	0.273	0.796			0.371	0.262	0.11	0.42	0.641	0.301	0.346	
9	0.415	0.472	-0.313	0.246	0.322	0.378	0.383	0.319	0.903	0.457	0.506	0.335	0.265	0.347	0.386	0.411		

Note: PC (perceived control)= 1, AMB (AI manipulation benefit)= 2, AMC (AI manipulation cost)= 3, INV (content involvement)= 4, LOY (platform loyalty)= 5, PP (perceived personalization)= 6, PK (persuasion knowledge)= 7, PR (perceived responsiveness)= 8, AMI (AI manipulation intention)= 9.

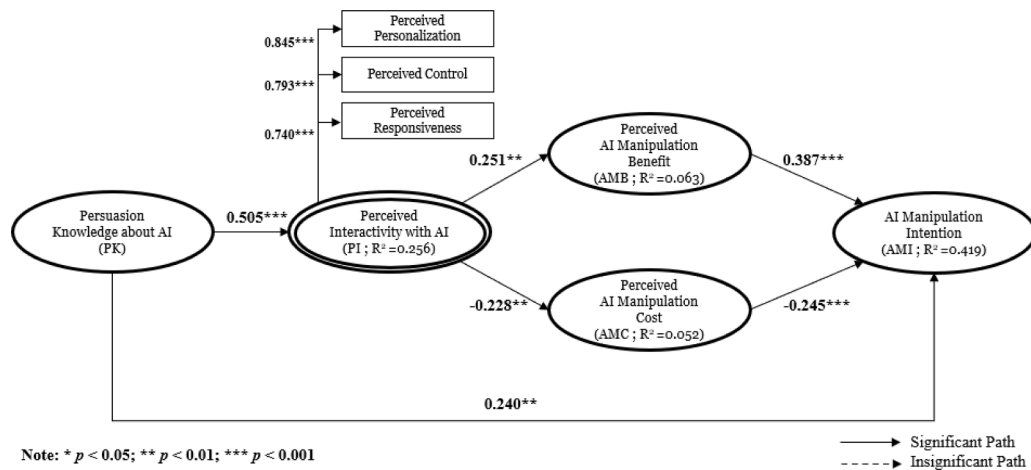


Fig. 6. Main structural equation modeling.

Table 6

Mediation analysis.

Indirect path	β Indirect effect	t-value	95 % confidence interval
PK → PI → AMB	0.127	2.768**	(0.035, 0.215)
PK → PI → AMB → AMI	0.049	2.209*	(0.012, 0.099)
PI → AMB → AMI	0.097	2.350*	(0.024, 0.185)
PK → PI → AMC	-0.115	2.692**	(-0.194, -0.026)
PK → PI → AMC → AMI	0.028	2.052*	(0.007, 0.060)
PI → AMC → AMI	0.056	2.079*	(0.012, 0.117)

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. PK (persuasion knowledge), PI (perceived interactivity), AMB (AI manipulation benefit), AMC (AI manipulation cost), AMI (AI manipulation intention).

Table 7

Path comparison.

Comparison	Path	Path coefficients	Standard error	t-value
AMB vs. AMC	AMB → AMI	0.387	0.069	104.748***
	AMC → AMI	-0.245	0.061	
AMB vs. PK	AMB → AMI	0.387	0.069	22.338***
	PK → AMI	0.240	0.073	
AMC vs. PK	AMC → AMI	-0.245	0.061	-77.821*
	PK → AMI	0.240	0.073	

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. PK (persuasion knowledge), AMB (AI manipulation benefit), AMC (AI manipulation cost), AMI (AI manipulation intention).

Table 8

Post-hoc test of MANOVA.

Dependent variable	Persuasion knowledge level		Levene's test (p-value)	Mean difference (SD)		
	Level	Mean (SD)		Medium vs. low	High vs. medium	High vs. low
CLICK	Low	3.93 (1.264)	0.480 (0.619)	0.30 (0.194)	0.45 (0.202)	0.75*** (0.181)
	Medium	4.23 (1.071)				
	High	4.68 (1.157)				
LIKE	Low	3.62 (1.254)	2.314 (0.101)	0.07 (0.231)	0.79** (0.240)	0.86*** (0.215)
	Medium	3.69 (1.385)				
	High	4.47 (1.585)				
DISLIKE	Low	3.05 (1.315)	2.950 (0.054)	0.37 (0.226)	0.15 (0.235)	0.52* (0.211)
	Medium	3.43 (1.217)				
	High	3.58 (1.550)				
FOLLOW	Low	3.64 (1.115)	1.484 (0.229)	0.23 (0.202)	0.57* (0.210)	0.80*** (0.188)
	Medium	3.87 (1.258)				
	High	4.44 (1.335)				

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; all tests were two-tailed. CLICK (intention to click on more content), DISLIKE (intention to press “dislike” for more content), FOLLOW (intention to subscribe to or follow more content uploaders), LIKE (intention to press “like” for more content); SD is short for standard deviation.

5.6. Additional analysis for intentions to engage in specific AI manipulation behaviors

To clarify whether we can treat CLICK, LIKE, DISLIKE, and FOLLOW as intentions to engage in specific AI manipulation behaviors, we further investigated how overall AI manipulation intentions related to intentions to engage in specific CLICK, LIKE, DISLIKE, and FOLLOW behaviors. We first incorporated the four variables into our research model. Users must click on the focal content before they can like or dislike it or follow its creator on YouTube and Instagram, so our model must consider that user actions are sequential. For this reason, we anticipate that CLICK influences LIKE, DISLIKE, and FOLLOW choices.

Fig. 7 shows the additional structural equation modeling results. All hypotheses were supported and consistent with the main results. In addition, CLICK positively and significantly influenced LIKE, DISLIKE, and FOLLOW, as expected. AI manipulation intention was also positively associated with CLICK, LIKE, and FOLLOW, but it was non-significantly related to DISLIKE. We examined the discrepancy by observing how AI manipulation intention affected DISLIKE through CLICK. We found that CLICK fully mediated the impact of AI manipulation intention on DISLIKE ($\beta = 0.142$, p -value = 0.003). Similarly, CLICK positively and partially mediated impacts of manipulation intentions on LIKE ($\beta = 0.146$, p -value = 0.002) and FOLLOW ($\beta = 0.157$, p -value < 0.001). These findings suggest that, as expected, manipulation intention indirectly influences intentions to engage in disliking, liking, and following behavior through the initial intentions of clicking on content.

In summary, additional analyses indicated that CLICK, LIKE, DISLIKE, and FOLLOW are AI manipulation intentions. Furthermore,

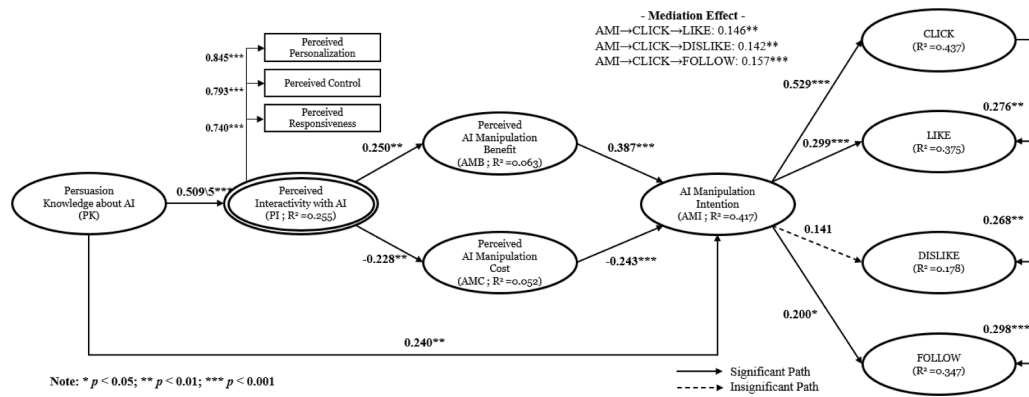


Fig. 7. Additional structural equation modeling.

our findings show that we need to consider sequential user actions and clicking as a mediator in the relationship between overall AI manipulation intentions and intentions to engage in liking, disliking, and following behaviors.

6. Discussion

Despite growing interest in AI-human interactions, the AI manipulation behavior literature has mostly concentrated on outputs and processes, without considering antecedents. To fill the gap, in this study, we identify antecedents that drive users to intend to manipulate AI. We also show how user knowledge levels affect manipulation intentions. We report results of survey data collected from 233 YouTube or Instagram users. PLS-SEM analysis supports our proposed hypotheses. Furthermore, MANOVA reveals that PK affects intentions to engage in different AI manipulation behaviors.

We find that PK regarding AI directly affects AI manipulation intentions and indirectly impacts AI manipulation intentions through perceived interactivity with AI and perceived costs/benefits from AI manipulation. Our findings align with AI manipulation research indicating that knowledge about AI relates to manipulation intentions (e.g., [3,11,33]). Hence, we conclude that PK is a primary antecedent motivating individuals to intend to manipulate AI.

This study shows that AI manipulation intentions are positively related with manipulation benefits and negatively related with manipulation costs. Moreover, path comparison tests indicate that perceived benefits have the greatest effect on AI manipulation intentions, followed by PK and costs. The results support economics studies arguing that cost-benefit assessments precede AI manipulation behaviors [8,33,34].

We find that perceived interactivity with AI is positively associated with perceived benefits and negatively associated with perceived costs. Our additional mediation analysis shows that perceived interactivity with AI significantly and indirectly affects AI manipulation intentions. The findings confirm the importance of interactivity in the AI manipulation context, supporting contentions that interactivity is a major factor for adaptation [1,72–74] and AI usage behaviors [23].

By being the first to show that PK is positively and significantly related to perceived interactivity in the AI context, we extend the PKM literature. PK is known to impact perceived personalization or control [23,84,90,95], which are subdimensions of interactivity. The few studies investigating how interactivity relates to PK examine non-AI contexts [96,97]. Thus, we newly confirm the positive and significant relationship between PK and perceived interactivity in the AI context.

We find that users who have strong knowledge about AI intend to click on more content, to press the “like” or “dislike” buttons for more content and subscribe to or follow more content uploaders. Those actions suggest higher intentions to manipulate AI. PK was shown to increase objective behavioral responses to AI recommendation algorithms [23], but we need better understanding about whether intentions to

manipulate the algorithm underlie PK’s positive impacts on the use of recommendations. In this sense, we extend research by investigating how PK relates to four types of AI manipulation intentions.

Although our findings may raise concerns about unethical AI misuse, understanding AI manipulation behavior remains essential, regardless of potential positive or negative impacts. On one hand, AI manipulation behavior should be promoted if it positively affects firms and individuals. On the other hand, if it is harmful, we must identify the factors that encourage such behavior before developing effective countermeasures.

6.1. Theoretical implications

Our work makes several theoretical contributions. First, we contribute to the information systems (IS) field and the literature on AI-human interaction by providing a systematic literature review on AI manipulation and developing a theoretical framework to explain AI manipulation behaviors. We review and synthesize the large but fragmented literature regarding AI manipulation behaviors, identifying four major conceptions of AI manipulation behaviors and recognizing the lack of studies regarding antecedents. For example, AI manipulation behavior studies have taken an economic perspective and mostly focused on mathematical models (e.g., [4,8]). Despite increasing interest in AI manipulation, as far as we can ascertain, the antecedents have been neglected. Therefore, we contribute to the IS and AI-human interaction literature by synthesizing the fragmented literature on AI manipulation, conceptualizing AI manipulation, and developing a comprehensive theoretical framework to understand AI manipulation intentions.

Second, we extend the PKM literature by examining how PK impacts interactivity and AI manipulation intentions. Specifically, most PKM studies focus on advertising. Only a few apply the PKM to AI user behaviors (e.g., [23]). AI is an interactive system [100,101], and interactivity is the key factor influencing AI decision-making processes [102]. Nevertheless, PKM research on AI has primarily explored how PK relates to perceived personalization or control [90,95]. Some studies investigate how PK relates to interactivity perceptions but in non-AI contexts [96,97]. Thus, researchers have yet to examine how perceived PK relates to interactivity in AI contexts. We fill this gap by finding that PK regarding AI is positively associated with perceived interactivity, including personalization, control, and responsiveness. We also find that PK directly and indirectly influences intentions to manipulate AI.

Third, this study extends prior literature on adaptation and misuse behavior. In this research, AI manipulation behavior can be considered as adaptation or misuse behavior in the AI context because it involves intentional adaptive AI usage or misuse of AI to achieve specific outcomes, which may deviate from the intended purposes or ethical guidelines. Most of the extant literature on adaption and misuse

behavior has focused on non-AI contexts, and few studies have investigated them in the AI context (e.g., [69,75]). Thus, by examining the antecedents of AI manipulation intention, we make novel contributions to the growing adaptation and misuse behavior literature.

6.2. Practical implications

This research has several practical implications for the various platforms where digital content is recommended, including social media and over-the-top media services.

First, managers who wish to encourage users to intend to manipulate AI should try to increase users' PK and perceived interactivity. We find that PK and perceived interactivity have significant direct and indirect effects on perceptions of benefits, costs, and intentions to manipulate AI. Our extended analysis shows that higher PK strengthens four types of AI manipulation intentions. Users are most likely to intend to manipulate AI when they perceive that it will be beneficial, and that PK and manipulation costs are secondary considerations. Therefore, managers should thoroughly explain how AI works (e.g., Wang and Benbasat [132]), make AI more personalized, inform users that their behaviors determine AI outcomes, and make AI's interface more interactive. On the contrary, to discourage AI manipulation intention, managers might reduce user interaction, for example, by slowing the AI update cycle.

Second, management should note that PK is positively associated with perceived interactivity, including the three subdimensions of personalization, control, and responsiveness. Perceived interactivity has been positively associated with service reuse intention [133] and platform trust [111]. Therefore, managers are advised to increase user knowledge regarding AI's mechanism to increase interactivity with AI. Users who have high interactivity perceive that AI is more customized to their needs, that they have better control over AI, and that AI responds more quickly to their input.

Third, our findings have implications for AI system design. Specifically, the results of the MANOVA and post hoc tests show that liking content and following or subscribing to content uploaders are more advanced AI manipulation techniques than clicking on and disliking content. Additional analysis shows that clicking acts as a significant mediator between AI manipulation intentions and liking, disliking, and following intentions, considering that user actions are sequential. Thus, practitioners must carefully interpret clicking-on-content signals because users may do so to intend to manipulate AI rather than to view the content.

6.3. Limitations and future research directions

This study has three major limitations. First, our survey-based approach carries inherent limitations regarding internal validity. For example, the PK we assess may inaccurately reflect the actual PK. Although our survey provides detailed explanations and attention checks, some respondents might misunderstand how intentional AI manipulation behaviors differ from basic clicking behaviors. In addition, we advise cautious interpretation of the findings regarding intentions to participate in CLICK, LIKE, DISLIKE, and FOLLOW types of AI manipulation behaviors, because they may misrepresent real-world behaviors. Similarly, people may express an intention to manipulate AI, but this does not necessarily translate into behavior. Various factors, such as situational constraints [134,135], can affect whether intentions are acted upon. Hence, future experiments should definitively verify the causal relationship between PK and AI manipulation behavior. Moreover, future researchers could analyze clickstream log data from real users or collect qualitative data from think-aloud protocols and interviews to understand how and why users manipulate AI.

The second limitation is that users of platforms other than YouTube and Instagram might respond differently to the survey questions. For example, e-commerce websites like Amazon differ from social media platforms in using AI recommendation algorithms. Consequently, users

may show different AI manipulation behaviors on e-commerce websites. Although we focus only on AI recommendation algorithms, AI manipulation behavior appears in other contexts such as ChatGPT where users have more complex text-based interactions with AI. They might try to change outcomes or show that they are dissatisfied with the previous responses by providing additional questions, instructions, examples, and data, or explicitly expressing likes or dislikes for ChatGPT results [136]. Our work provides some insights about ChatGPT manipulation, but we expect that ChatGPT manipulation behaviors are more diverse and complicated because of the text-based interactions. Therefore, it would be valuable to examine the mechanisms behind e-commerce websites and AI manipulations among ChatGPT users.

Third, our focus is on settings where users benefit from AI manipulation, rather than environments where AI manipulation can harm individuals or algorithms. Therefore, future research should explore antecedents of AI manipulation behaviors in other settings.

CRedit authorship contribution statement

Taeyoung Kim: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Il Im:** Writing – review & editing, Writing – original draft, Validation, Project administration, Investigation, Funding acquisition, Conceptualization.

Acknowledgements

This work was supported by the 'BK21 FOUR (Fostering Outstanding Universities for Research)' in 2023. The authors are sincerely grateful to the editor and the anonymous reviewers for their invaluable comments. We would also like to express our gratitude to the professors and students of the Information Systems Area at Yonsei University for their valuable feedback.

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Taeyoung Kim is a PhD candidate at the School of Business, Yonsei University. His research interests include AI-human interaction and the economics of IS. He has published his papers in the Americas Conference on Information Systems (AMCIS), the Korean Chapter of the Association for Information Systems (KraIS) workshop, and the Workshop on e-Business (WeB).

Il Im is a professor of information systems at the School of Business, Yonsei University. He received his Ph.D. from Marshall School of Business, University of Southern California. Before joining Yonsei University, he was an assistant professor in the Information Systems Department at the New Jersey Institute of Technology. His current research focuses on personalization technologies and their effects, technology acceptance, and platform strategy. His research has been published in journals such as *MIS Quarterly*, *ACM Transactions on Information Systems*, *Information & Management*, *Decision Support Systems*, *Computers in Human Behavior*, and *Journal of Business Research*.